CHEMISTRY ONLY

15 EXTRACTION AND USES OF METALS

Metals are some of the most important materials that we use in everyday life. This chapter explores the principles behind the extraction of metals from their ores. We will also look at the properties of alloys and the uses of some metals.

LEARNING OBJECTIVES

- Know that most metals are extracted from ores found in the Earth's crust and that unreactive metals are often found as the uncombined element
- Explain how the method of extraction of a metal is related to its position in the reactivity series, illustrated by carbon extraction for iron and electrolysis for aluminium
- Be able to comment on a metal extraction process, given appropriate information *(detailed knowledge of the processes used in the extraction of a specific metal is not required)*
- Explain the uses of aluminium, copper, iron and steel in terms of their properties *(the types of steel will be limited to low-carbon (mild), high-carbon and stainless)*
- Know that an alloy is a mixture of a metal and one or more elements, usually other metals or carbon
- Explain why alloys are harder than pure metals

EXTRACTING METALS FROM THEIR ORES

MINERALS AND ORES

Most metals are found in the Earth's crust combined with other elements. The individual compounds are called **minerals**.



▲ Figure 15.1 Pyrite (iron pyrites), FeS_2



▲ Figure 15.2 Magnetite, Fe_3O_4

DID YOU KNOW?

The price of a metal is affected by how common the ore is and how difficult it is to extract the metal from the ore.

Figures 15.1 and 15.2 show samples of some iron-containing minerals; they are normally found mixed with other unwanted minerals in rocks. An **ore** is a sample of rock that contains enough of a mineral for it to be worthwhile to extract the metal. Most metals are extracted from ores found in the Earth's crust.

HINT

You do not need to remember the names of these ores/minerals for the exam.

A few very unreactive metals, such as gold, are found **native**. That means that they exist naturally as the uncombined element. Silver and copper are also sometimes found native, although much more rarely.

EXTRACTING THE METAL

Many ores contain either oxides or compounds that are easily converted to oxides. Sulfides such as sphalerite (zinc blende), ZnS, can be easily converted into an oxide by heating in air, a process known as **roasting**.

$$2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$$

REMINDER

If you have forgotten about oxidation and reduction, you might find it useful to re-read Chapter 14.

To obtain the metal from the oxide, you have to remove the oxygen. Removal of oxygen is called reduction. Metals exist as positive ions in their ionic compounds, and to produce the metal you would have to add electrons to the positive ion. Addition of electrons is also called reduction.

METHODS OF EXTRACTION AND THE REACTIVITY SERIES

How a metal is extracted depends to a large extent on its position in the reactivity series. A manufacturer obviously wants to use the cheapest possible method of extracting a metal from an ore. There are two main economic factors to take into account:

- the cost of energy
- the cost of the reducing agent.

potassium
sodium
calcium
magnesium
aluminium
(carbon)
zinc
iron
copper

▲ Figure 15.3 A part of the reactivity series

METALS BELOW CARBON IN THE REACTIVITY SERIES

For a metal below carbon in the reactivity series, the cheapest method of reducing the ore is often to heat it with carbon.

The extraction of iron is a good example of this. One of the main ores of iron contains a high percentage of iron(III) oxide. The iron can be extracted from this by heating with carbon:

$$Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(l) + 3CO(g)$$

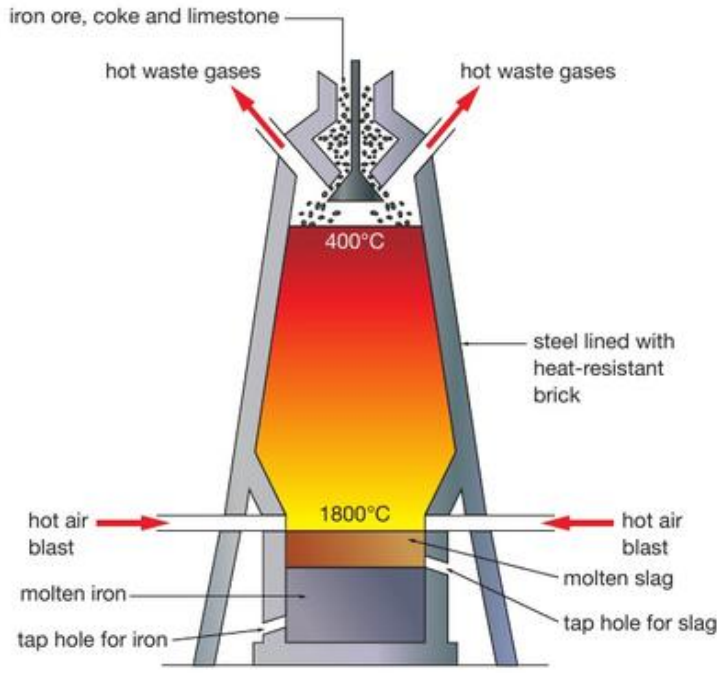
Carbon is higher in the reactivity series than iron and will take the oxygen away from the iron oxide. This is a redox reaction; the Fe_2O_3 is reduced to Fe in the reaction and the C is oxidised to CO. In this reaction the carbon is the reducing agent, it reduces the iron(III) oxide.

The extraction of iron is carried out in a blast furnace. The process is a bit more complicated than we have shown here. Other reactions also occur and the main reducing agent is actually carbon monoxide:

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$$

HINT

You are not required to learn the details of the blast furnace.



▲ Figure 15.4 Iron is extracted from its ore in a blast furnace.

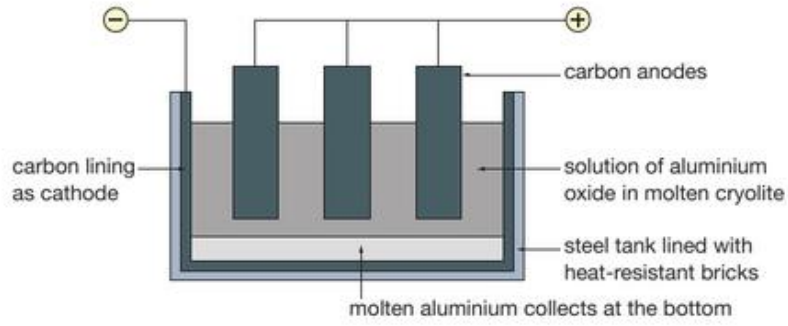
METALS ABOVE CARBON IN THE REACTIVITY SERIES

Ores of metals higher in the reactivity series than zinc can't be reduced using carbon at reasonable temperatures. This is because the metals are more reactive than carbon and therefore carbon cannot take the oxygen away from the metal oxide. Metals above zinc are usually produced by *electrolysis*.

KEY POINT

This is essentially the same as the electrolysis of molten aluminium oxide. The cryolite is simply used to make the process more economical. Aluminium oxide melts at over 2000 °C but if it is dissolved in molten cryolite the whole process can be carried out at around 1000 °C. This saves a lot of energy and therefore also saves a lot of money.

Aluminium is extracted by the electrolysis of aluminium oxide (Al_2O_3) dissolved in a molten salt called cryolite.



▲ Figure 15.5 Aluminium is produced by electrolysis.

At the cathode the reaction is $Al^{3+} + 3e^- \rightarrow Al$ reduction

At the anode the reaction is $2O^{2-} \rightarrow O_2 + 4e^-$ oxidation

This process requires huge amounts of electricity, which makes it expensive. A metal such as aluminium, which has to be extracted by electrolysis, is much more expensive than one like iron, which can be extracted by reduction with carbon, despite the fact that aluminium is more abundant in the Earth's crust than iron. Iron (and all other metals) could also be extracted by electrolysis but because the process is so expensive we do not use it unless we have to.

Some metals, such as titanium, are extracted by heating the compound with a more reactive metal. This is also an expensive method because the more reactive metal itself will have had to be extracted by an expensive process first.

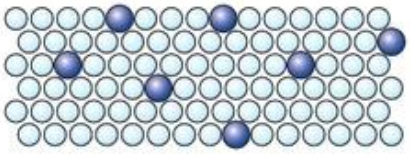
ALLOYS



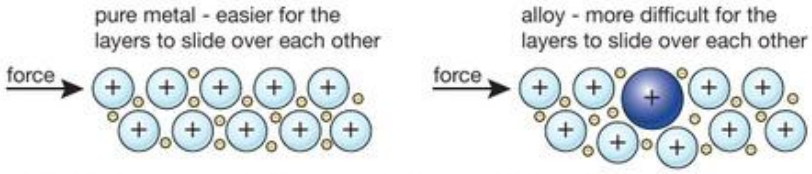
▲ Figure 15.6 A brass propeller. Brass is an alloy.

An **alloy** *is a mixture of a metal with, usually, other metals or carbon*. For example, brass is a mixture of copper and zinc, and steel is an alloy of iron with carbon.

Alloys are harder than the individual pure metals from which they are made. In an alloy, the different metals/elements have slightly differently sized atoms. This breaks up the regular lattice arrangement and makes it more difficult for the layers of ions to slide over each other.



▲ Figure 15.7 Atoms in an alloy



▲ Figure 15.8 The layers cannot slide over each other as easily in an alloy as in a pure metal.

REMINDER

Remember, a metal consists of a regular lattice of metal ions surrounded by a sea of delocalised electrons.

Other common alloys include bronze (a mixture of copper and tin), stainless steel (an alloy of iron with chromium and usually nickel), and the mixture of copper and nickel (cupronickel) which is usually used to make 'silver' coins.

PROPERTIES AND USES OF SOME METALS

USES OF ALUMINIUM

Pure aluminium isn't very strong, so aluminium alloys are normally used instead. The aluminium can be strengthened by adding other elements, such as silicon, copper or magnesium.

Figure 15.9 shows one of the uses of aluminium.



▲ Figure 15.9 Aluminium resists corrosion, it has a low density and is a good conductor of electricity. The aluminium in the cables in the photo is strengthened by a core of steel.

Other uses of aluminium include planes, electricity cables, and pots and pans. Its uses depend on its low density and strength (when alloyed), its ability to conduct electricity and heat, and its ability to resist corrosion.

Aluminium resists corrosion because it has a very thin, but very strong, layer of aluminium oxide on the surface. This prevents anything else reaching the surface and reacting with it.

PROPERTIES AND USES OF DIFFERENT KINDS OF STEEL

There are lots of different alloys of iron that contain various proportions of carbon and sometimes other metals. An alloy of iron and carbon is called steel.

MILD STEEL

Mild steel is the name given to an alloy of iron containing up to about 0.25% of carbon. This small amount of carbon increases the hardness and strength of the iron. Mild steel is a strong material that can be easily hammered into various shapes (malleable) and drawn into wires (ductile). It is used for (among other things) nails, car bodies, ship building, girders and bridges.



▲ Figure 15.10 Mild steel is used for car bodies . . .

A disadvantage of mild steel is that it rusts when exposed to oxygen and water. It is also about three times denser than aluminium. Some car bodies are made from aluminium. This has the advantage that the car body will not rust and, because the car is lighter, less fuel will have to be used.



▲ Figure 15.11 Some cars have a body made of aluminium. In this photo the car body has been painted. The paint is mainly for decoration; it is not required to prevent rusting.

HIGH-CARBON STEEL

High-carbon steel is iron containing about 0.6–1.2% carbon (these amounts are variable). High-carbon steel is harder and more resistant to wear than mild steel but more brittle (not as malleable and ductile). It is used for cutting tools. High-carbon steel also usually contains small amounts of manganese.



▲ Figure 15.12 High-carbon steel is used to make cutting tools.

STAINLESS STEEL



▲ Figure 15.13 Some cars have a stainless steel body. They do not need to be painted.

Stainless steel is an alloy of iron with chromium and often nickel. Chromium forms a strong oxide layer in the same way as aluminium, and this oxide layer protects the iron as well. Stainless steel is therefore very resistant to corrosion.

Uses include kitchen sinks, saucepans, knives and forks, and gardening tools, but there are also major uses for it in the brewing (making beer), dairy (milk and cheese production) and chemical industries, where corrosion-resistant vessels are essential. Stainless steel is significantly more expensive than mild steel.

Table 15.1 A summary of types of steel

Type of steel	Iron mixed with	Some uses
mild steel	up to 0.25% carbon	nails, car bodies, ship building, girders
high-carbon steel	0.6–1.2% carbon	cutting tools, masonry nails
stainless steel	chromium (and nickel)	cutlery, cooking utensils, kitchen sinks

COPPER



▲ Figure 15.14 Copper pots and pans

Some properties and uses of copper and its alloys are summarised in Table 15.2.

Table 15.2 Copper and its alloys have a variety of uses

Use	Property
electrical wires	very good conductor of electricity and ductile
pots and pans	very good conductor of heat (thermal conductor), very unreactive and malleable
water pipes	unreactive – does not react with hot or cold water and malleable
surfaces in hospitals	antimicrobial properties and malleable

CHAPTER QUESTIONS

SKILLS REASONING, PROBLEM SOLVING 9

1 Explain why different methods are used to extract aluminium and iron from their ores and write equations for the reactions occurring.

SKILLS CREATIVITY 10

2 Sodium is the sixth most abundant element in the Earth's crust, occurring in large quantities as common salt, NaCl, and yet sodium metal wasn't first produced until the early 19th century.

a From your knowledge of the position of sodium in the reactivity series, suggest a method for manufacturing sodium from sodium chloride. You aren't expected to give details of the manufacturing process, but should describe and explain (including equation(s) where relevant) how sodium is formed in your process.

SKILLS REASONING 7

b Suggest why sodium wasn't produced until the early 19th century.

c Suggest three other metals which might have been first isolated from their compounds at the same sort of time.

SKILLS CREATIVITY (8)

3 Lead is between iron and copper in the reactivity series. Explain a method that could be used to extract lead from its ore.

4 The extraction of chromium from its ore is quite complicated, but in the final stage chromium(III) oxide is reacted with aluminium to form chromium and aluminium oxide.

SKILLS PROBLEM SOLVING (9)

a Write a balanced equation for this reaction.

SKILLS REASONING

b Explain what the reducing agent is in this reaction.

(8)

c Explain whether chromium is more or less reactive than aluminium.

5 The first stage in the extraction of titanium from its ore is heating the ore with chlorine and carbon. The equation for the reaction is:

$$2FeTiO_3 + 7Cl_2 + C \rightarrow 2TiCl_4 + 2FeCl_3 + CO$$

This equation is not completely balanced.

The $TiCl_4$ is separated from the $FeCl_3$ and then, in the second stage, heated with magnesium.

SKILLS PROBLEM SOLVING

a Balance the equation for the first stage in this process.

(9)

b Write a balanced chemical equation for the second stage in the process.

SKILLS REASONING (8)

c Explain whether magnesium is more or less reactive than titanium.

SKILLS CREATIVITY

d Explain a method that could be used to extract magnesium from its ore.

SKILLS REASONING

e Explain whether you would expect titanium to be more or less expensive than magnesium.

SKILLS CRITICAL THINKING (4)

6 This question is about the properties and uses of some metals.

a Aluminium alloys are used in aircraft construction.

i Explain which property of aluminium makes it particularly suitable for this purpose.

SKILLS REASONING (8)

ii Explain, in terms of structure and bonding, why aluminium alloys are used in preference to pure aluminium.

(7)

iii Some cars have aluminium bodies rather than mild steel bodies. Explain two advantages and one disadvantage of using aluminium rather than mild steel.

b Explain two properties of copper that make it suitable for use in:

(4)

i electrical wiring

(5)

ii water pipes

(7)

c Explain why car bodies are made from mild steel rather than high-carbon steel.

END OF CHEMISTRY ONLY